Function: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Function, domain, co-domain, range

KEY POINTS

- $f : A \to B$: every element of A has exactly one image in B
- $f(a) = b$: b image of a, a pre-image of b
- Domain A, co-domain B, range $= \{f(a)\} \subseteq B$
- Uniform: one rule; non-uniform: piecewise rules
- Vertical line test: a vertical line meets a function's graph at most once

2. Domain rules

Formulas

$$D_{f\pm g} = D_{fg} = D_f \cap D_g, \qquad D_{f/g} = D_f \cap D_g - \{g(x) = 0\}$$

KEY POINTS

- Denominator $\neq 0$; even root: radicand ≥ 0
- $\log_b a$: $a > 0$, $b > 0$, $b \neq 1$
- $\sin^{-1} u, \cos^{-1} u$: $|u| \leq 1$; ${}^nC_r, {}^nP_r$: $n \in N$, $r \in W$, $r \leq n$
- $a^{p/q}$: q even needs $a \geq 0$; $f^g = e^{g \ln f}$ needs $f > 0$
- $\max / \min(f, g)$: $D_f \cap D_g$

Watch out

Never simplify before finding the domain: $\dfrac{x^2 - 4}{x - 2}$ has domain $R - \{2\}$

3. Standard functions

KEY POINTS

- $\sin, \cos$: $R \to [-1, 1]$; $\tan$: $R - \{(2n + 1)\frac{\pi}{2}\} \to R$; $\cot$: $R - \{n\pi\} \to R$
- $\sec, \operatorname{cosec}$: range $(-\infty, -1] \cup [1, \infty)$
- a^x: $R \to (0, \infty)$; $\log_a x$: $(0, \infty) \to R$; mirror images in $y = x$
- Polynomial: domain R; odd degree: range R, at least one real zero; even degree: range never R
- Rational g/h: $R - \{h = 0\}$; algebraic vs transcendental (trig, exp, log)
- $x + \frac{1}{x}$: range $(-\infty, -2] \cup [2, \infty)$; $x - \frac{1}{x}$: range R

STEPS

1. Exponential equation: same base, or $t = a^x > 0$
2. Log equation: combine logs, remove log, check roots in the domain

4. Modulus, signum

Formulas

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \qquad \operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

KEY POINTS

- $|x|$: $R \to [0, \infty)$; sgn: $R \to \{-1, 0, 1\}$
- $x \operatorname{sgn} x = |x|$, $|x| \operatorname{sgn} x = x$, $\operatorname{sgn}(\operatorname{sgn} x) = \operatorname{sgn} x$
- $|f(x)|$: reflect the part below the x-axis upward

5. Greatest integer and fractional part

Formulas

$$[x] \le x < [x] + 1, \qquad \{x\} = x - [x] \in [0, 1)$$

$$[x] + [-x] = \begin{cases} 0, & x \in I \\ -1, & x \notin I \end{cases} \qquad \{x\} + \{-x\} = \begin{cases} 0, & x \in I \\ 1, & x \notin I \end{cases}$$

KEY POINTS

- $[x]$: $R \to I$; $\{x\}$: $R \to [0, 1)$, period 1
- $[x + m] = [x] + m$, $\{x + m\} = \{x\}$ for $m \in I$
- $[x] + [y] \le [x + y] \le [x] + [y] + 1$; $\{[x]\} = [\{x\}] = 0$
- $[f(x)]$, $\{f(x)\}$: split where $f(x)$ crosses integers

STEPS

1. $[f] = k$: $k \in I$ and $k \le f < k + 1$
2. $[f] \ge k \iff f \ge k$; $[f] > k \iff f \ge k + 1$; $[f] < k \iff f < k$ $(k \in I)$
3. Mixed equation: put $x = n + f$, $0 \le f < 1$, test each integer n

Watch out

$[-2.7] = -3$, not -2; $\{-3.7\} = 0.3$

6. Range

STEPS

1. Solve $x = g(y)$; range $= \{y : g(y)$ real and in $D_f\}$
2. Quadratic in x: $D \ge 0$ (check leading coefficient $= 0$ separately)
3. Composite: range of inner $= T$; range $= f(T)$ (monotonic: end values)
4. Non-monotonic on an interval: include turning-point values
5. Piecewise: union of the ranges of the pieces

KEY POINTS

- $a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$; $A^2 + B^2 \ge 2AB$ for $\sin^2 + \operatorname{cosec}^2$ type

Watch out

Remove y-values produced only by excluded x (cancelled factors)

7. Identical functions, special functions

KEY POINTS

- Identical: same domain and same values (graphs coincide)
- Identity $I_A(x) = x$ (bijection); constant $f = c$ (range $\{c\}$)
- Bounded: $|f(x)| \le M$; homogeneous of degree n: $f(tx, ty) = t^n f(x, y)$
- Implicit to explicit: solve the relation for y, state the domain

Watch out

$\ln e^x \ne e^{\ln x}$ as functions (domains differ)

8. Transformation of graphs

KEY POINTS

- $f(x) + b$: up/down; $f(x + a)$: left $(a > 0)$/right
- $f(ax)$: x-coordinates divided by a; $af(x)$: y-values times a
- $-f(x)$: mirror in x-axis; $f(-x)$: mirror in y-axis
- $|f(x)|$: lower part flipped up; $f(|x|)$: right part copied to the left
- $f(-|x|)$: left part copied to the right; $|y| = f(x)$: upper part copied below
- $\dfrac{ax + b}{cx + d}$: hyperbola, centre $\left(-\frac{d}{c}, \frac{a}{c}\right)$

Watch out

Order matters: $|f(|x|)|$ = first $f(|x|)$, then modulus

9. One-one and many-one

Formulas

$$\text{one-one} : f(a) = f(b) \Rightarrow a = b$$

STEPS

1. Graph: every horizontal line cuts at most once
2. Derivative: $f' \geq 0$ (or ≤ 0) throughout, zero only at isolated points
3. Cubic: one-one $\iff b^2 \leq 3ac$

KEY POINTS

- One-one: linear, $\frac{ax+b}{cx+d}$, a^x, $\log_a x$, x^3, $x|x|$
- Many-one: even degree polynomials, Q/L, Q/Q, trig on full domain, $|x|$, $[x]$, $\{x\}$, even, periodic
- Rational/irrational rules: each rule one-one and no common image

10. Onto, into, bijective

KEY POINTS

- Onto: range = co-domain; into: range $\subsetneq$ co-domain
- Onto/into depends on the co-domain
- $R \to R$ polynomial: odd degree onto; even degree neither onto nor one-one
- Bijective = one-one + onto = invertible

Watch out

Always compare the range with the given co-domain, not with R

11. Counting functions ($n(A) = m$, $n(B) = n$)

Formulas

$$\text{Total} = n^m, \quad \text{One-one} = {}^nP_m \ (m \leq n), \quad \text{Onto} = \sum_{r=0}^{n} (-1)^r \, {}^nC_r (n-r)^m \ (m \geq n)$$

KEY POINTS

- $m = n$: one-one = onto = $n!$; many-one = into = $n^n - n!$
- $m > n$: one-one 0; $m < n$: onto 0
- Many-one = total − one-one; into = total − onto

STEPS

1. With conditions: fix the restricted elements first, then the rest
2. Strictly increasing maps: ${}^nC_m$; non-decreasing: ${}^{n+m-1}C_m$

12. Functional equations

STEPS

1. Type 1: $f(g(x)) = h(x)$: put $g(x) = t$
2. Type 2: $f(x)$ with $f(\frac{1}{x})$, $f(1-x)$, $f(-x)$: replace x, solve two equations
3. Type 3: odd part + constant: use $f(x) + f(-x)$
4. Special values: put $x = 0$, $y = 0$, $x = y$, $y = -x$

Formulas

$$f(x+y) = f(x)+f(y) \Rightarrow f(x) = kx; \quad f(x+y) = f(x)f(y) \Rightarrow f(x) = a^x; \quad f(xy) = f(x)+f(y) \Rightarrow f(x) = k \log x$$

$$f(x)f\left(\tfrac{1}{x}\right) = f(x) + f\left(\tfrac{1}{x}\right) \Rightarrow f(x) = 1 \pm x^n$$

KEY POINTS

- Polynomial with $f(i) = g(i)$: $f - g = k(x-1)\cdots(x-n)$
- $f(x) + f(1-x) = c \Rightarrow \sum_{k=1}^{n-1} f\left(\frac{k}{n}\right) = \frac{(n-1)c}{2}$

13. Composite functions

Formulas

$$(g \circ f)(x) = g(f(x)), \qquad D_{g\circ f} = \{x \in D_f : f(x) \in D_g\}$$

KEY POINTS

- Not commutative; associative
- $f \circ g$ one-one $\Rightarrow g$ one-one; $f \circ g$ onto $\Rightarrow f$ onto
- Composite of two bijections is a bijection
- f^n: look for a cycle $f^k = I$; linear $px + q$: $f^n = p^n x + q\frac{p^n - 1}{p-1}$

STEPS

1. Non-uniform: write $f(g(x))$ by pieces of f, solve $g(x) \in$ each piece from the graph of g

14. Even and odd functions

Formulas

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

KEY POINTS

- Even: $f(-x) = f(x)$, symmetric about y-axis; odd: $f(-x) = -f(x)$, symmetric about O
- Domain must be symmetric; only $f \equiv 0$ is both
- Derivative: odd $\to$ even, even $\to$ odd
- $E \pm E = E$, $O \pm O = O$, $E \cdot E = O \cdot O = E$, $E \cdot O = O$
- Composite: even if either is even; odd if both odd
- Standard odd: $\ln\frac{1-x}{1+x}$, $\ln(x + \sqrt{1+x^2})$, $\frac{a^x - 1}{a^x + 1}$

15. Inverse function

Formulas

$$f(f^{-1}(x)) = x, \quad f^{-1}(f(x)) = x, \quad (g \circ f)^{-1} = f^{-1} \circ g^{-1}, \quad (f^{-1})^{-1} = f$$

STEPS

1. Check bijective (restrict domain / co-domain if needed)

2. $y = f(x)$, solve for x, swap x and y, pick the branch by the range

KEY POINTS

- $D_{f^{-1}} = R_f$, $R_{f^{-1}} = D_f$; graphs mirror in $y = x$
- Increasing f: $f(x) = f^{-1}(x) \iff f(x) = x$
- $\dfrac{ax+b}{cx+d}$ is self-inverse iff $a + d = 0$

16. Periodic functions

Formulas

$$f(x+T) = f(x), \qquad T_{f(ax+b)} = \frac{T}{|a|}, \qquad \text{LCM}\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{\text{LCM}(a,c)}{\text{HCF}(b,d)}$$

KEY POINTS

- $\sin, \cos, \sec, \text{cosec}$: 2π; $\tan, \cot$, moduli: π; even powers: π
- $\{x\}$: 1; $|\sin x| + |\cos x|$: $\frac{\pi}{2}$; constant: no fundamental period
- af, $f + c$, $1/f$: same period
- Sum/product: LCM; rational with irrational: aperiodic
- Bounded domain or non-repeating gaps: aperiodic

STEPS

1. Find each period, take LCM, then test $\frac{LCM}{2}$ etc.
2. Functional equation: shift x until $f(x+T) = f(x)$

17. Common mistakes checklist

STEPS

1. Simplified before finding the domain?
2. Log base > 0 and $\neq 1$ checked?
3. Onto checked against the given co-domain?
4. Strict or non-strict end in $[f(x)]$ inequalities?
5. Checked half of the LCM for the period?
6. Inverse branch chosen by the range?
7. Composite domain: range of inner inside domain of outer?